A393172

A Catalan substitution congruence modulo four

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AI disclosure. ChatGPT (OpenAI) generated the mathematical arguments, Lean proofs, and this manuscript in response to Dev Goyal's requests. Dev Goyal is the sole human author. The source sequence and conjecture or observation are credited to Paul D. Hanna. **Independent human review is pending.**

Abstract

The unique normalized integral solution of $A(x - x^2) = x + A^2$ satisfies $A \equiv 2C - x \pmod 4$, where $C = x + C^2$. Hence its coefficients above degree one have residue two precisely at powers of two, and zero otherwise. The comparison satisfies the literal source equation.

1 Source, notation, and scope

The source is Paul D. Hanna's OEIS entry A393172 [1], retrieved on 24 October 2026. Its initial sequence index is 1. We also retain the formal zero constant coefficient at index zero.

All series are formal. The set $\mathbb{N}$ includes zero, and $[x^n]U$ denotes a coefficient. A normalized series is $x + O(x^2)$; $O(x^r)$ means divisibility by x^r, over any coefficient ring used below. We write $U \circ V = U(V(x))$ only when $V(0) = 0$. Compositional inversion, denoted $\mathrm{Rev}(U)$ or $U^{\langle -1 \rangle}$, is distinct from multiplicative inversion of a unit. Congruences are coefficientwise; ordinary products and compositional iterates are distinct.

2 Formal algebra used in the proof

Lemma 2.1 (Triangular construction)**.** *Let K be a commutative ring. Suppose Φ preserves normalized series and improves agreement by one degree:*

$$G - H = O(x^r) \quad \Longrightarrow \quad \Phi(G) - \Phi(H) = O(x^{r+1}) \qquad (r \geq 2).$$

Then Φ has a unique normalized fixed point. The same conclusion holds for a family of series if the degree-n equations, at every family index, determine the degree-n coefficients from strictly lower degrees.

Proof. The degree-n coefficient of $\Phi(G)$ depends only on coefficients of G below degree n. Fix the constant and linear coefficients, and then define every later coefficient successively. This gives a fixed point; the same induction proves uniqueness. For a family, perform the induction on degree simultaneously at every family index. The same induction applies with only the constant coefficient prescribed, provided agreement is improved starting at degree one. It also applies to an operator on all series when agreement is improved starting at degree zero. For the

operators used in this paper, the recurrence requires only ring operations, so it also constructs integral solutions when $K = \mathbb{Z}$. Equivalently, iterating Φ from any normalized initial series fixes one further coefficient at each step. □

Lemma 2.2 (Inversion and transport)**.** *A normalized series over any commutative ring has a unique normalized two-sided compositional inverse. A constant-one series has a unique multiplicative inverse with constant coefficient one. These inverses and valid composition commute with coefficient ring homomorphisms. Normalized compositional inversion preserves agreement modulo x^r.*

Proof. For normalized G, the degree-n equation in $G(H) = x$ determines h_n with multiplier one from earlier coefficients. Construct a right inverse J of H. Associativity gives $G = G \circ (H \circ J) = (G \circ H) \circ J = J$, so $H \circ G = x$ as well. The same associativity argument proves uniqueness. Multiplicative inversion follows similarly from the coefficient recurrence for $UV = 1$.

Each coefficient of a valid composition is a finite polynomial in the coefficients of its arguments. Ring homomorphisms preserve these polynomials and the defining equations, so uniqueness gives transport of inversion. If $G - H = O(x^r)$, composition with any normalized series preserves that agreement. Composing with the respective inverses then shows $\mathrm{Rev}(G) - \mathrm{Rev}(H) = O(x^r)$. □

Lemma 2.3 (Binary series and doubling)**.** *There is a unique normalized integral series $C = x + C^2$. Its reduction modulo two is*

$$P(x) = \sum_{k \geq 0} x^{2^k}.$$

For series U, V over $\mathbb{Z}/4\mathbb{Z}$, the equality $2U = 2V$ is equivalent to equality of their reductions modulo two.

Proof. The operator $G \mapsto x + G^2$ improves agreement because $G^2 - H^2 = (G - H)(G + H)$ and normalized $G + H$ has zero constant coefficient. In characteristic two, $U(x)^2 = U(x^2)$ for every formal series. Thus the normalized equation $U = x + U^2$ forces the coefficient at one to be one, every higher odd-index coefficient to be zero, and the coefficient at $2m$ to equal the coefficient at m. This gives exactly P.

For scalar residues modulo four, $2u = 2v$ holds precisely when u and v have the same parity. Apply this statement coefficientwise. In particular, this argument does not cancel a factor of two in a ring in which two is a zero divisor. □

3 Source construction and the modulus-four theorem

Let $C = x + C^2$ and put $N = x - x^2$. Then $N \circ C = x$, and Lemma 2.2 also gives $C \circ N = x$. We use A for the source sequence A393172 [1].

Lemma 3.1 (A useful square equation)**.** *For normalized $T, P \in K[[x]]$, there is a unique normalized solution of*

$$G = T + (G \circ P)^2.$$

The construction commutes with coefficient ring homomorphisms.

Proof. If $G - H = O(x^r)$, then $(G - H) \circ P = O(x^r)$. The sum $G \circ P + H \circ P$ has zero constant coefficient, so the difference of the two squares is $O(x^{r+1})$. Normalization is preserved. Lemma 2.1 gives existence and uniqueness; mapping the equation and using uniqueness gives transport. □

Theorem 3.2 (A393172 modulo four)**.** *The unique normalized integral solution of*

$$A \circ N = x + A^2 \tag{1}$$

has, for every $n > 1$*,*

$$[x^n]A \equiv \begin{cases} 2 \pmod 4, & n = 2^k \text{ for some } k \geq 0, \\ 0 \pmod 4, & \text{otherwise.} \end{cases}$$

Proof. Composing (1) with C gives $A = C + (A \circ C)^2$. Conversely, composing this equation with N gives (1). Lemma 3.1 therefore proves integral existence and uniqueness for the literal source equation.

Over $\mathbb{Z}/4\mathbb{Z}$, the normalized candidate $D = 2C - x$ satisfies

$$D \circ N = 2x - N = x + x^2, \qquad D^2 = x^2 + 4(C^2 - xC) = x^2.$$

Thus D solves (1), and uniqueness gives $A = 2C - x$ modulo four. For $n > 1$, the coefficient of x contributes nothing and Lemma 2.3 shows that twice the coefficient of C is two precisely at powers of two, and zero elsewhere. □

4 Separate Lean proof and checked scope

The entry file is `lean/Entries/A393172.lean`. Its explicit theorem statements and proof terms use the namespace `FinalCollection.A393172`. The complete imported implementations and their shared mathematical dependencies are included in the collection. The following local theorem names give the proof map:

- `first_existence`.
- `first_definition`.
- `first_residue`.
- `first_arbitrary_source`.
- `complete_series_mod_four`.
- `divisibility_complement`.

The main implementation modules are

`Round4.CatalanComposition`

`Round4.SquareComposition`

The project pins Lean 4.19.0 and Mathlib `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The supplied collection report records fresh source compilation and transitive axiom audits. All coefficient conclusions are unbounded.

Limits and status. Normalized source equation $A(x - x^2) = x + A(x)^2$ is constructed uniquely. The standard Catalan parity mechanism is not claimed as a new general result. No global novelty, exclusive priority, peer review, or OEIS acceptance is asserted.

References

[1] Paul D. Hanna, *OEIS A393172*, 4 February 2026. `https://oeis.org/A393172`. Retrieved 24 October 2026.